1. Introduction: Intentional Spacecraft–Debris Interactions in a Crowded Orbit

The current low-Earth orbit (LEO) environment is increasingly populated with objects, with more deliberate interactions between them and a growing use of automation to manage those interactions. Most discussions of collision risk still focus on random close approaches between satellites and debris, but that lens misses an important change: commercial operators are now intentionally flying close to other objects for servicing, inspection, and active debris removal (ADR).

Intentionality fundamentally changes both risk and responsibility. Rather than only avoiding debris, operators now deliberately place spacecraft into close proximity with non-cooperative objects, introducing new forms of operational uncertainty and potential failure.

This paper focuses on Intentional U.S. Private Spacecraft ↔ Debris interactions, defined as deliberate maneuvers by a U.S.-licensed commercial spacecraft to approach and interact with orbital debris that has no identifiable active owner. As ADR concepts move from demonstration to routine operations, these interactions will shape both the physical debris environment and the evolution of U.S. governance.

To explore this, the paper develops a hypothetical failure scenario, maps the current governance framework across hard law, industry standards, and self-governance, examines key operational sub-types, and identifies prevention and liability gaps.

2. Hypothetical Scenario: A U.S. Servicer Meets Unowned Debris

Imagine Orion Dynamics, a U.S.-based startup focused on commercial active debris removal. Its flagship spacecraft, OD-1, is a dedicated ADR servicer designed to rendezvous with and de-orbit large debris objects in congested low Earth orbit (LEO). The vehicle is equipped with a robotic arm, close-proximity sensors (LIDAR and optical cameras), and a de-orbit kit capable of either executing a controlled reentry burn or attaching a drag device.

The OD-1 mission is modeled on real-world ADR programs that have moved from concept to on-orbit demonstration. Astroscale's ELSA-d mission (2021) demonstrated magnetic capture of a client satellite in LEO, and its ELSA-M program now targets multi-client commercial debris removal (Astroscale Holdings, 2021). Other relevant demonstrations include ClearSpace-1's planned capture of a Vega rocket adapter and Northrop Grumman's MEV docking operations with active Intelsat satellites. Collectively, these programs illustrate the transition of ADR from theory to operational reality, and the governance gaps that follow (European Space Agency, 2023; Northrop Grumman, n.d.).

Orion obtains the necessary U.S. authorizations, including launch, reentry, and communications approvals, and submits a debris mitigation plan detailing its approach to capturing an upper stage left in orbit decades earlier. The object is cataloged debris, passivated, and has no active operator, making it effectively “unowned” for the purposes of this analysis.

After launch, OD-1 performs phasing maneuvers to match the target’s orbit. It then conducts intentional proximity operations, closing from kilometers to tens of meters while synchronizing with the object’s slow tumbling motion. These maneuvers rely on a combination of onboard autonomy and ground supervision, with human operators setting constraints and autonomous guidance, navigation, and control (GNC) systems handling fine positioning.

In the final approach, OD-1 enters capture mode. The robotic arm extends as the spacecraft stabilizes for contact with a structural feature on the upper stage. At this stage, the mission’s risk is highest: the servicer is deliberately operating in close proximity to a non-cooperative, tumbling object with uncertain dynamics.

A combination of factors turns the maneuver into an accident. A small error in estimating the target's tumbling rate, compounded by a brief sensor dropout, leads to a misaligned arm motion. Instead of a clean capture, the arm strikes a fragile section of the upper stage, imparting unexpected torque and fragmenting part of the structure. OD-1 sustains damage and aborts the operation.

In the aftermath, debris fragments disperse along the object's orbit. While most are small, several are trackable. One fragment later intersects the orbital path of a third-party commercial imaging satellite. Because its trajectory change is not captured in time, it collides with the satellite, rendering it inoperable.

This sequence illustrates how an intentional U.S. ADR maneuver targeting unowned debris can damage the servicer itself and create debris that might subsequently destroy a third-party satellite, raising critical questions about authorization, due care, liability, and acceptable risk in ADR operations.

3. Governance Map: Hard Law, Industry Standards, and Self-Governance

The OD-1 failure scenario highlights how intentional spacecraft–debris interactions are shaped by multiple layers of governance rather than a single, unified regime. At one end are hard-law obligations, including international treaties and U.S. domestic regulation, that set broad requirements for authorization, supervision, and liability. In the middle are industry standards and technical guidelines, which influence behavior but do not carry direct legal force. At the other end are company-level self-governance choices, where operators define internal risk thresholds, autonomy policies, and safety culture. Together, these three tiers form the framework within which OD-1 and similar ADR missions are conceived, licensed, and conducted.

3.1. Hard Law: Treaties and Domestic Regulation

At the international level, the mission engages obligations under the Outer Space Treaty and the Liability Convention. Under the treaty's authorization and continuing supervision requirement, the United States must authorize Orion Dynamics' activities and oversee OD-1's conduct (Outer Space Treaty, art. VI). The treaty's due regard and harmful interference principles also apply, requiring that intentional proximity operations be conducted without creating unreasonable risks to other operators (Outer Space Treaty, art. IX).

The Liability Convention becomes relevant once OD-1's failed maneuver generates debris that damages a third-party satellite. In orbit, liability is fault-based, raising the question of whether the United States, as the launching state, could be held responsible if the servicer's conduct falls below a standard of due care (Liability Convention, art. III).

At the domestic level, Orion must navigate a patchwork of U.S. regulatory authorities. Launch, reentry, and communications are licensed and reviewed for safety, while Orbital Debris Mitigation Standard Practices (ODMSP) guide debris-related risk assessments (14 C.F.R. Part 450; U.S. Government Orbital Debris Mitigation Standard Practices, 2019). However, these frameworks were developed primarily for traditional satellite operations and do not yet provide detailed, binding criteria for high-risk ADR proximity operations against unowned debris.

3.2. Industry Standards: Formal Guidelines Without Legal Force

Between hard law and internal decision-making lies a growing body of industry standards and technical guidelines. These outline practices for limiting debris creation, planning disposal, and conducting rendezvous and proximity operations (RPO), including approach corridors, safety buffers, and coordination protocols (IADC Space Debris Mitigation Guidelines, 2012; UNCOPUOS Long-term Sustainability Guidelines, 2018).

In the OD-1 scenario, such standards influence mission design but lack legal enforceability. They may recommend risk assessments, minimum separation distances, and coordination with space situational awareness (SSA) providers. However, there is no universally binding rule specifying abort criteria, acceptable collision probabilities during ADR operations, or required coordination when interacting with unowned debris.

3.3. Self-Governance: Company-Level Risk Choices

The final tier is self-governance, which is particularly significant in ADR operations. Orion Dynamics sets its own internal risk thresholds, determining acceptable probabilities of capture failure, the balance between autonomy and human control, and how aggressively to pursue uncertain capture opportunities.

These choices live in the company's day‑to‑day practices and software, not in formal law. In OD‑1's case, choosing an aggressive capture approach and not being conservative enough about the target's tumbling rate were self‑governance decisions that helped cause the accident. The fact that the mission aimed to clean up debris does not remove the need to ask whether the company's safety margins were strong enough.

Taken together, the OD-1 scenario demonstrates that intentional ADR missions are governed by a mix of broad treaty obligations, evolving domestic regulation, non-binding standards, and internal company decisions, with no single framework specifically tailored to intentional spacecraft–debris interactions.

4. Three Operational Sub-Types of Intentional Debris Interaction

While OD-1 represents a dedicated ADR servicer, intentional U.S. spacecraft–to–debris interactions can take multiple forms. In addition to dedicated servicers, two other categories are relevant:

- **Constellation-based ADR capabilities:** Large satellite networks performing distributed, often opportunistic debris interactions, increasing regulatory complexity due to scale and autonomy.
- **Dual-use and ASAT-adjacent systems:** Debris-removal platforms with the ability to approach or interfere with active satellites, raising security, transparency, and strategic concerns.

Together, these categories illustrate that intentional debris interactions are not a single use case but a range of operational models with distinct governance challenges.

5. Gaps and Implications

The OD-1 scenario and its extensions across different operational models reveal key gaps in the current governance landscape for intentional debris interactions.

5.1. Prevention Gaps

Existing hard-law frameworks provide only high-level obligations. Neither the Outer Space Treaty nor the Liability Convention defines what constitutes adequate due care for ADR proximity operations involving unowned, non-cooperative debris. U.S. regulations and the Orbital Debris Mitigation Standard Practices (ODMSP) introduce mitigation expectations but do not establish detailed, binding criteria for high-risk debris interactions.

In particular, there is no standardized framework specifying:

- safe approach geometries or minimum standoff distances for debris capture;
- abort thresholds based on sensor uncertainty or tumbling-rate estimation;
- mandatory coordination or notification procedures for deliberate high-risk proximity operations.

Industry guidelines, including the Inter-Agency Space Debris Coordination Committee (IADC) guidelines, begin to address these issues but remain voluntary and unevenly adopted. (IADC Space Debris Mitigation Guidelines, 2012) As a result, responsibility for defining "safe enough" behavior is largely delegated to operators' internal processes. For instance, if a binding standard had required Orion to abort the capture attempt when sensor uncertainty exceeded X% or when tumbling-rate estimation error surpassed Y threshold, the collision might have been avoided. No such standard exists, so Orion's self-governance decision to proceed became the only operative rule. As ADR missions scale, especially with greater autonomy, this lack of a shared baseline risks inconsistent practices and increased collision risk.

The limitations of the ODMSP are significant. The ODMSP, first published in 1997 and updated periodically, establishes general best practices for minimizing debris generation, including limiting orbital lifetimes, passivating propulsion and energy systems at end of life, and avoiding intentional break-ups (U.S. Government Orbital Debris Mitigation Standard Practices, 2019). However, the ODMSP was designed for conventional satellite operations and makes no specific provision for active proximity operations with debris, capture attempts, or the controlled approach of a servicer to a non-cooperative tumbling target. More broadly, it does not address how to manage the risks of intentionally interacting with someone else's defunct hardware. This reflects the fact that commercial ADR was not a realistic scenario when the framework was developed. As the U.S. government updates its orbital debris policies, a process ongoing through the Office of Space Commerce and the interagency National Space Council, developing criteria tailored to ADR proximity operations should be a priority (Space Policy Directive-3, 2018).

5.2. Liability and Accountability Gaps

On the liability side, the OD-1 scenario highlights a central tension: how to assign responsibility when a socially beneficial but inherently risky ADR mission causes harm. Current frameworks provide general principles of fault and state responsibility but do not clearly distinguish between negligent conduct and high-risk remediation activities undertaken in good faith.

If an operator follows existing guidelines yet still generates debris that damages a third-party satellite, it is unclear how liability should be allocated. Strict fault-based approaches may discourage ADR operations, while more permissive interpretations risk normalizing unsafe behavior.

This challenge becomes more complex across operational models. In constellation-based ADR, responsibility may be distributed across many small interactions, complicating causal attribution. In dual-use systems, accidents may carry not only commercial and environmental consequences but also strategic and diplomatic implications. Existing legal frameworks do not provide tailored liability rules for these distinct forms of intentional debris interaction.

5.3. Nationality Variation: How Foreign Ownership Changes the Analysis

The nationality of the damaged third-party asset materially alters the legal and diplomatic landscape. In the baseline OD-1 scenario, if the imaging satellite struck by the secondary debris is operated by a U.S. company, the dispute remains within the domestic legal system, and compensation would likely be resolved through insurance, commercial arbitration, or U.S. tort law. The Liability Convention would not be triggered as an inter-state claim because both parties are U.S. nationals.

If, however, the damaged satellite belongs to a foreign operator, say, a European or Japanese commercial imaging company, the Liability Convention's fault-based framework for damage in space becomes directly relevant as an inter-state claim. Under Article III of the Liability Convention, the launching state of the damaging object (here, the United States as the state that authorized and supervised OD-1) bears international liability if the damage resulted from its fault or the fault of persons for whom it is responsible. (Liability

Convention, art. III) The U.S. government would face a formal diplomatic claim from the foreign state and would need to assess whether Orion's conduct met the standard of due care, a standard that, as discussed above, remains undefined for ADR proximity operations. This inter-state exposure creates a strong incentive for the U.S. to develop clearer domestic licensing criteria before commercial ADR operations proliferate.

A distinct issue arises when the debris object itself has a foreign origin. If the upper stage that OD-1 targeted was originally launched by a non-U.S. state, questions arise about whether the originating state retains any residual obligations with respect to that object, for example, under the Outer Space Treaty's registration and continuing supervision requirements, and whether it could share liability for the consequences of failed removal attempts. No international framework currently resolves these questions clearly, exposing ADR operators to legal uncertainty depending on the provenance of the debris they target.

5.4. Autonomy Variation: How the Degree of Automation Changes the Analysis

The OD-1 hypothetical assumes a hybrid autonomy model: human operators set high-level parameters and authorize major burns, while onboard GNC algorithms manage fine relative motion and last-meter positioning. This blend represents the current state of practice for most commercial proximity operations. However, the governance analysis shifts considerably depending on where the human-machine boundary is drawn, both in the direction of less autonomy and more.

Where a human operator retains positive control throughout the final approach and capture attempt, reviewing sensor data in real time and commanding each maneuver step, fault attribution is relatively straightforward. The operator's judgment calls become the proximate cause of any accident, and the standard of due care can be assessed against what a reasonable, trained human pilot should have done under the circumstances. Liability, under both the Liability Convention and potential domestic tort claims, maps reasonably well onto this model because there is an identifiable human decision-maker at each critical juncture. (Liability Convention, art. III)

Fully autonomous ADR operations where the spacecraft decides independently when to initiate capture, whether to abort, and how to respond to unexpected debris dynamics, introduce a fundamentally different accountability structure. If the accident in the OD-1 scenario had been caused entirely by the spacecraft's onboard algorithm without any human intervention, it becomes far less clear who bears responsibility: the engineer who designed the GNC software, the company that set the algorithm's risk parameters, or the operator who authorized deployment with those parameters in place. Current U.S. regulatory frameworks and international treaties were not designed with full machine autonomy in mind and provide no clear mechanism for assigning fault to algorithmic decision-making. As ADR servicers and constellation-based systems increasingly delegate proximity decisions to software, this gap will become one of the most pressing governance challenges in the field.

5.5. Implications for U.S. Policy

These gaps suggest that U.S. policy must evolve across all three tiers of the governance spectrum. At the hard-law level, regulators may need clearer licensing criteria and conditions specific to ADR missions, particularly for high-risk proximity operations and autonomous decision-making. At the industry standards level, there is a need for more concrete, technically grounded norms defining acceptable risk thresholds and safe operating practices. At the self-governance level, companies must develop and, where possible, disclose internal safety standards, risk tolerances, and autonomy frameworks commensurate with the risks they introduce.

As orbital congestion increases and intentional, autonomous interactions become more common, the stakes of effective governance will rise. The OD-1 scenario underscores that intentionality is a double-edged sword: it enables proactive debris remediation while simultaneously introducing new pathways for accidents and misinterpretation. Addressing this tension will be a central challenge for U.S. space policy in the coming decade.

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